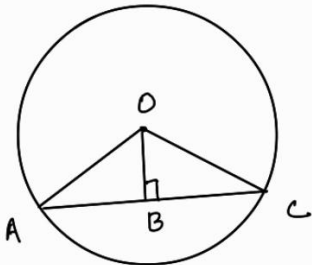
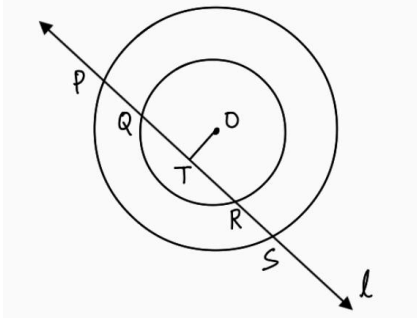




GRADE: IX	Sample Paper PA- 2	YEAR: 2026-27
TIME: 180 Min	SUBJECT: MATHEMATICS (041)	MAX. MARKS: 80

Q No.	SECTION - A		Marks (20X1=20)
1.	Find the zeroes of $x^2 - (p + q)x + pq$		1M
	a. p, q	b. $-p, q$	
	c. $-p, -q$	d. $p, -q$	
2.	The sum of two positive numbers is 30 and their product is 176. The positive difference between them is:		1M
	a. 10	b. 14	
	c. 7	d. 24	
3.	A boy runs 800 m on day 1 and increases his distance by 70 m daily. How far does he run on the 4th day?		1M
	a. 828 m	b. 1028 m	
	c. 8280 m	d. 1010 m	

4.	One end of the diameter of a circle is $(-2, 5)$. Find the coordinates of the other end of it, if the centre of the circle is $(2, -1)$.		1M
	a. $(-7, 6)$	b. $(4, -7)$	
	c. $(5, 6)$	d. $(6, -7)$	
5.	Which of the following is integer number:		1M
	a. $(2+2\sqrt{2}) + (-2+2\sqrt{2})$	b. $(2+2\sqrt{2}) - (4-2\sqrt{2})$	
	c. $(2-\sqrt{2})(2-2\sqrt{2})$	d. $(2+2\sqrt{2})(2-2\sqrt{2})$	
6.	What is the value of AC if the radius of the circle is 5 cm and $OB = 3$ cm? 		1M
	a. 5 cm	b. 8 cm	
	c. 4 cm	d. 2 cm	
7.	If the length of the diagonal of the square is $5\sqrt{2}$ cm. Then what is the perimeter of the square?		1M
	a. 25 cm	b. 15 cm	
	c. 30 cm	d. 20 cm	

8.	A sector of a circle has radius 21 cm. The ratio of the major arc to the minor arc is 5:3. The length of major arc is:		1M
	a. 26.25π cm	b. 48π cm	
	c. 45.5π cm	d. 52.5π cm	
9.	What is the ratio of PQ/RS if OT is perpendicular to line l?		1M
			
	a. 0.5	b. 2	
	c. 1.5	d. 1	
10.	If $a=5/8$ and $b=15/24$ then how many rational numbers between them:		1M
	a. Exactly one rational number	b. Exactly two rational numbers	
	c. No rational number	d. Infinitely many rational numbers	
11.	The distance of the point (-9,-12) from origin will be		1M
	a. 13	b. 15	
	c. 14	d. 17	

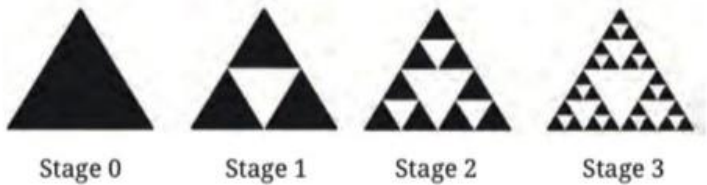
12.	The first term of a GP is 15, and the 4th term of this GP is 3240. What will be the ratio of the 8th term to the 6th term of this GP?		1M
	a. 36	b. 24	
	c. 25	d. 49	
13.	A square has side length $(x + 5)$ cm, its area is 169 cm^2 . Find x ?		1M
	a. 6	b. 7	
	c. 8	d. 9	
14.	If the equation of a line is $3x + 2y = 6$ then x-intercept is _____ and y-intercept is _____		1M
	a. 3, 2	b. 2, 3	
	c. 2, 6	d. 3, 6	
15.	Which of the following equations represents a linear growth relationship.		1M
	a. $y = 5x - 2$	b. $y = x^2 + 3$	
	c. $y = 2^3 + 4$	d. $y = x^3 + 1$	
16.	Evaluate: $(3x + 2)^2 - (3x - 2)^2$		1M
	a. $6x$	b. $24x$	
	c. 0	d. $9x$	

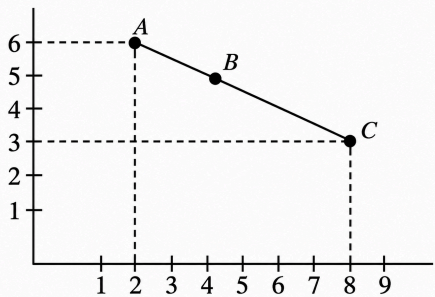
17.	A series is given with one term missing. Select the correct alternative from the given ones that will complete the series. 213, 221, ? , 240, 251, 263, 276, 290		1M
	a. 231	b. 233	
	c. 227	d. 230	
18.	If three distinct points are collinear, then the area of triangle formed by them is:		1M
	a. 1 sq unit	b. 0 sq unit	
	c. Depends on coordinates	d. Cannot be determined	
19.	Assertion(A): The decimal expansion of $7/40$ is terminating. Reason(R): The denominator $40 = 2^3 \times 5$, which has no prime factors other than 2 and 5.		1M
	a. Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).	b. Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).	
	c. Assertion (A) is true but reason (R) is false.	d. Assertion (A) is false but reason (R) is true.	
20.	Assertion(A): If two chords are at the same distance from the centre , then they are equal. Reason(R): Equal chords are always parallel.		1M

	a. Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).	b. Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).	
	c. Assertion (A) is true but reason (R) is false.	d. Assertion (A) is false but reason (R) is true.	
	SECTION B (Very Short Answer Type Questions)		(5X2=10)
21.	The perimeter of a circle is 44 cm. What is its radius?		2M
22.	An auto-rickshaw fare starts at ₹25 and remains the same for initial 2 km. Then it increases by ₹15 per km. What will be the fare for a trip of 10 km?		2M
	OR		
22.	Suppose the length of the rectangle is 13 cm. Find the area, if breadth is (i) 12 cm, (ii) 10 cm. Find the linear pattern representing the area of the rectangle.		2M
23.	Plot the points A(-1, 2), B(2, 2), C(2, -1) and D(-1, -1) in the coordinate plane. Is ABCD a square? Justify		2M
24.	Represent $\sqrt{5}$ on the number line.		2M
	OR		

24.	Prove that $\sqrt{3}$ is an irrational number.	2M
25.	Factorise the following: $8x^3 + 36x^2y + 54xy^2 + + 27y^3$	2M
	SECTION C (Short Answer Type Questions)	(6X3=18)
26.	Prove that chords of a circle having the same length are all at the same distance from the centre of the circle.	3M
27.	A staircase has 30 steps. The height of the first step is 12 cm, and each successive step is 0.4 cm higher than the previous one. What is the total height (in meters) reached after climbing all 30 steps?	3M
28.	Find the perimeter of the sector of a circle of radius 14 cm and angle 45° .	3M
	OR	
28.	A sector is cut from a circle of radius 21 cm. The angle of the sector is 150° . Find the length of its minor arc and major arc?	3M
29.	Convert into p/q from: 0.23232323.....	3M
30.	The village playground is shaped as a square of side 40 m. A path of width s meters is created around the playground for people to walk. Find an	3M

	expression for the area of the path in terms of s .	
31.	If $(1,2)$, $(4,y)$, $(x,6)$ and $(3,5)$ are the vertices of parallelogram taken in order. Find x and y .	3M
	OR	
31.	If the mid-point of the line segment joining the points $A(3,4)$ and $B(k,6)$ is $P(x,y)$ and $x + y - 10 = 0$. Find the value of k .	3M
	SECTION D (Long Answer Type Questions)	(4X5=20)
32.	A park is being developed in the shape of a sector of a circle. A decorative fence is to be installed only along the boundary of the sector. The radius of the sector is 21 m and the central angle is 120° . (i) Find the length of the fencing required to enclose the park. (ii) A 2 m wide entrance gate is left unfenced on one of the radii. Find the total length of fencing actually required and its cost at ₹450 metre.	5M
33.	In a circle, if the distance of chord AB from centre is twice the distance of another chord CD from the centre, then can we conclude that $CD = 2AB$? Give reasons for your answer.	5M
	OR	
33.	When two chords intersect, each of them is divided into two line segments. Show that if the intersecting chords are of equal length, then	5M

	the line segments of one chord are equal to corresponding line segments of the other chord.	
34.	Consider the relationship between temperature measured in degrees Celsius($^{\circ}\text{C}$) and degrees Fahrenheit($^{\circ}\text{F}$), which is given by $^{\circ}\text{C} = a ^{\circ}\text{F} + b$. Find a and b, given that ice melts at 0 degrees Celsius and 32 degree Fahrenheit, and water boils at 100 degree Celsius and 212 degrees Fahrenheit.	5M
35.	Saira has arranged a square of side x units, 8 rectangular strips of side x units and width 1 unit and 15 squares of side 1 unit to form a bigger rectangle. Find the length and breadth of the rectangle in terms of x.	5M
	OR	
35.	Simplify: $\frac{(a^2 - b^2)^3 + (b^2 - c^2)^3 + (c^2 - a^2)^3}{(a - b)^3 + (b - c)^3 + (c - a)^3}$	5M
	Section E (Case - Study)	(3X4=12)
36.	Observe the Sierpinski triangle and try to answer the following questions:  Stage 0 Stage 1 Stage 2 Stage 3	
(i)	How many black triangles are there in stage 0 to 3?	1M

(ii)	Can you predict the number of black triangles at stage 4 and stage 5?	1M
(iii)	Can you find a rule for the number of black triangles at the n^{th} stage?	2M
	OR	
(iii)	Suppose the area of the triangle (that is the black region) in stage 0 is 1 square unit. What is the area of the black region in stage 1,2 and 3?	2M
37.	Read the following text and answer the questions given below: A group of class IX students goes on a picnic during summer holidays. Three friends Aman, Bhupesh and Chahal are sliding on a slider. The position of three friends are shown by the points A, B and C on the slider. 	
(i)	Find the distance between A and C?	1M
(ii)	Is B the mid point of AC? Justify by finding the mid-point of AC	1M
(iii)	Find the slope of line segment AC?	

		2M
	OR	
(iii)	Find the point on the x-axis which is equidistant from A and B.	2M
38.	Ryan and his sister Emma are making square cupcakes, each with a side of 10 cm. Ryan decides to add a creative touch by frosting along one of the diagonals of a square cupcake. This sparks a mini math adventure as they try to figure out the length of the diagonal.	
(i)	What is the length of the diagonal?	1M
(ii)	What is the ratio of the side of the cupcake to the diagonal of the cupcake? Also tell what kind of decimal expansion it has?	1M
(iii)	Emma has a rectangular tray with length $26+2\sqrt{19}$ cm and breadth $26-2\sqrt{19}$ cm. Find how many cupcakes they can fit in the tray.	2M
	OR	
(iii)	Instead of square cupcakes they decided to make rectangular cupcakes of length $10+2\sqrt{3}$ cm and breadth $5+\sqrt{5}$ cm. Then what will be the area of 1 cupcake?	2M
	<i>"A Genius is often merely a talented person who has done all of his or her homework"</i>	